OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2022-23

Course Instance Code(s)	4BMS2, 4BS2, 1SPS1, 1OA2
Examinations	4th Science
Module Codes	MA4344
Module	Advanced Group Theory
Paper No.	1
External Examiner(s)	Prof C. Roney-Dougal
Internal Examiner(s)	Prof G. Pfeiffer *Dr James Cruickshank
Instructions	**Attempt ALL FOUR questions**. Each question carries 25 marks in total. Marks for individual parts of questions are indicated inline.
Duration	2 hours
No. of Pages	3 pages (including this cover page)
Discipline	Mathematics

Requirements

Release in exam venue		Yes
MCQ answer aheet	No	
Handout	No	
Formula and Tables*		Yes
Cambridge Tables 2nd Edition	No	
Other materials	No	
Graphic material in colour	No	

Q1. (a) Consider the permutation $\sigma = (1,2,3,4)(3,4,5,6)$. Write σ as product of disjoint cycles [**3 marks**]. Write σ as a product of transpositions [**3 marks**].

(b) What is the definition of the *alternating group* A_n [**3 marks**]? Write down the elements of A_4 [**3 marks**]. Show that A_4 has a normal subgroup of order 4 [**5 marks**].

(c) Prove that, for $n \geq 3$, A_n is generated by its 3-cycles [**8 marks**].

Q2. (a) Show that the cyclic group $\mathbb{Z}_{15}$ can be written as an internal direct product of subgroups H and K, where $|H| = 3$ and $|K| = 5$ [**8 marks**].

(b) State the *First Isomorphism Theorem* [**3 marks**]. Use the first isomorphism theorem to show that $\mathbb{R}/\mathbb{Z} \cong S^1$ where $\mathbb{R}$ denotes the additive group of real numbers, $\mathbb{Z}$ denotes the additive group of integers and S^1 is the multiplicative group of complex numbers of modulus 1 [**7 marks**].

(c) Describe all the group homomorphisms from $\mathbb{Z}_{21}$ to $\mathbb{Z}_{18}$ [**7 marks**].

Q3. (a) Prove the following special case of *Cauchy's Theorem* [**10 marks**].

> **Theorem.** Let G be a finite abelian group of order n. If p is a prime that divides n, then G contains an element of order p.

(Note that you are not allowed to assume the Fundamental Theorem of Finite Abelian groups for this part)

(b) State the *Fundamental Theorem of Finite Abelian Groups* [**3 marks**]. Describe, up to isomorphism, all the abelian groups of order 40 [**4 marks**].

(c) Give an example of an abelian group that is not isomorphic to a direct product of cyclic groups [**3 marks**]. You should prove that your example has the required property [**5 marks**].

Q4. (a) Write down the *class equation* for a finite group G and explain the meaning of each of the terms of the equation [**3 marks**]. Suppose that G is a finite group of order p^r where p is a prime and $r \geq 1$. Show that $|ZG| > 1$ [**6 marks**].

(b) Let G be a finite group and let p be a prime that divides $|G|$. What is a *Sylow p-subgroup of* G [**3 marks**]? Prove the following proposition [**7 marks**].

> **Proposition.** Suppose that P is a Sylow p-subgroup of G and that $x \in G$ has order a power of p. If $x^{-1}Px = P$ then $x \in P$.

(c) Suppose that G is a group of order 40. Prove that G is not a simple group [**6 marks**].